

DS5216: Artificial Intelligence (2024/25)

Programming Assignment 02: Player Tracking in Sports Videos

Registration Number: MPHIL251011

```
#Install YOLOv8
!pip install ultralytics opencv-python
```

```
from google.colab import drive
drive.mount('/content/drive')
```

Person detection and Tracking

```
#instead of detecting all objects, detect only person
import os
from ultralytics import YOLO

# Load model
model = YOLO("yolov8n.pt")

video_folder =
"/content/drive/MyDrive/Colab_Notebooks/AI_Assignment2/videos"

# Process every video in the folder
for video in os.listdir(video_folder):

    if video.endswith(".mp4"):

        video_path = os.path.join(video_folder, video)

        print(f"Processing {video}...")

        model.predict(
            source=video_path,
            classes=[0],          # person class only
            conf=0.4,
            save=True,
            project="/content/drive/MyDrive/Colab_Notebooks/AI_Assignmen
t2",
            name="YOLOv8n_Output",
            exist_ok=True
        )

print("Finished!")
```

Speed: 3.0ms preprocess, 9.7ms inference, 1.6ms postprocess per image at shape (1, 3, 384, 640)
Results saved to /content/drive/MyDrive/Colab_Notebooks/AI_Assignment2/Tracking_Output
Tracking cri1.mp4

Examples of Output of Player Detection and Tracking



Figure 1: Player (Cricket) tracking and Detection



Figure 2: Player (Football) tracking and Detection



Figure 3: Player (Rugby) Tracking and Detection

Pose estimation

```
from ultralytics import YOLO
import os

pose_model = YOLO("yolov8n-pose.pt")

video_folder =
"/content/drive/MyDrive/Colab_Notebooks/AI_Assignment2/videos"

for video in os.listdir(video_folder):

    if video.endswith(".mp4"):
```

```
video_path = os.path.join(video_folder, video)

print(f"Processing {video}...")

pose_model.predict(
    source=video_path,
    conf=0.4,
    save=True,
    project="/content/drive/MyDrive/Colab_Notebooks/AI_Assignmen
t2",
    name="YOLOv8nPose_Output",
    exist_ok=True
)

print("Finished!")
```

Examples of Output of Players' Pose



Figure 4: Player detection and Pose detection of Cricket Players

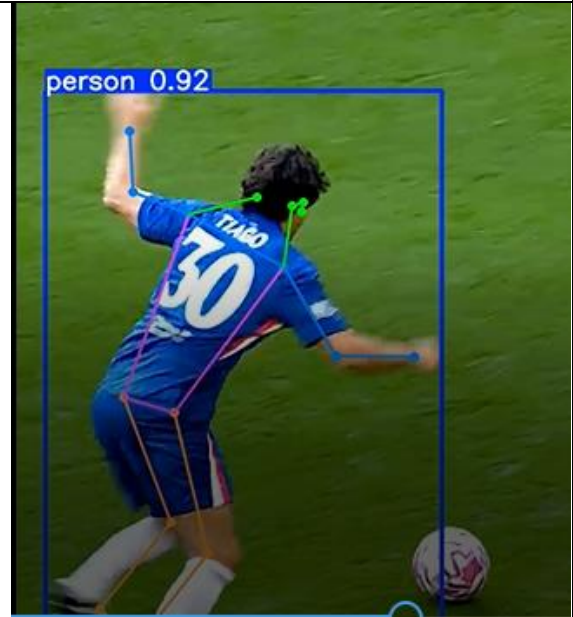


Figure 5: Player detection and Pose detection of Football Players

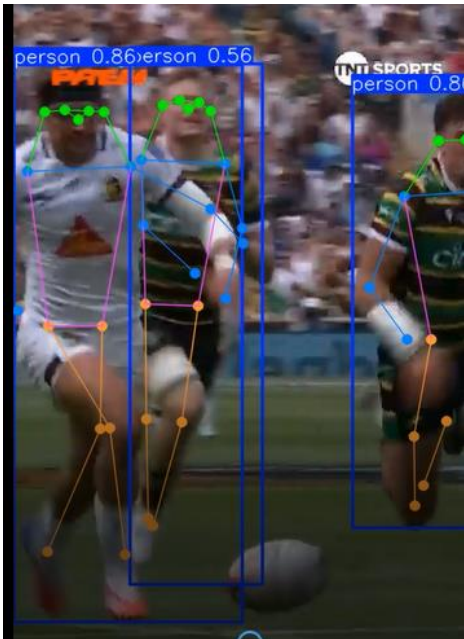


Figure 6: Player detection and Pose detection of Rugby Players

Performance comparison models

Since the pretrained YOLOv8n and YOLOv8n-Pose **models were evaluated on unlabeled sports videos** collected for this study, standard supervised evaluation metrics such as precision, recall, mAP, and training/validation loss curves could not be computed due to the absence of ground-truth annotations. Therefore, the performance of the models was evaluated based on computational metrics, including preprocessing time, inference time, postprocessing time, total processing time per frame, and qualitative assessment of player detection, player tracking, and key point detection across the six sports videos.

Table 1: Performance comparison models YOLOv8n and YOLOv8n-Pose

Metric	YOLOv8n (Tracking)	YOLOv8n-Pose
Preprocessing Time	3.7 ms	3.7 ms
Inference Time	11.1 ms	13.0 ms
Postprocessing Time	1.9 ms	1.8 ms
Total Time per Frame	16.7 ms	18.5 ms
Detects Players	Yes	Yes
Player Tracking	Yes	Yes
Body Keypoints	No	Yes

Conclusion

The results demonstrate a trade-off between speed and functionality. YOLOv8n achieved lower inference time (11.1 ms) and is therefore more suitable for applications where real-time processing is the main objective. In contrast, YOLOv8n-Pose required slightly longer inference time (13.0 ms) because it performs both player detection and body keypoint estimation. Although this introduces a modest computational overhead, the additional pose information significantly enhances the model's usefulness for sports analytics, player movement analysis, and action recognition. Therefore, YOLOv8n is recommended for efficient player detection and tracking, while YOLOv8n-Pose is preferable when detailed human pose information is required.

Discussion on Model Performance, Limitations, and Possible Improvements

The pretrained YOLOv8 models demonstrated effective real-time player detection and tracking across multiple sports videos. Among the evaluated models, YOLOv8n provided the fastest inference speed, making it suitable for real-time applications with limited computational resources. The YOLOv8n-Pose model further enhanced the analysis by estimating player body keypoints, enabling posture and movement analysis in addition to object detection. The generated tracking outputs showed that the models were capable of maintaining player identities across consecutive frames under normal playing conditions.

However, several limitations were observed during the experiments. Since the models were evaluated on custom sports videos without manually annotated ground-truth labels, standard evaluation metrics such as precision, recall, F1-score, and mean Average Precision (mAP) could not be computed. Consequently, the evaluation relied primarily on inference speed (FPS), processing time, and qualitative visual inspection of the tracking results. In addition, player occlusion, rapid movements, crowded scenes, and varying

camera angles occasionally resulted in missed detections, identity switches, or unstable tracking. Small or distant players were also more challenging to detect accurately, particularly in wide-angle recordings.

Several improvements can further enhance the system. Fine-tuning the pretrained YOLOv8 models using a labelled sports-specific dataset would significantly improve detection accuracy and robustness. Incorporating larger YOLOv8 variants (e.g., YOLOv8m or YOLOv8l) may increase detection performance, although at the expense of processing speed. Integrating advanced multi-object tracking algorithms such as ByteTrack or DeepSORT can improve identity consistency during occlusions and complex player interactions. Furthermore, expanding the evaluation using annotated datasets would enable quantitative assessment using standard performance metrics, providing a more comprehensive comparison of model performance.